GradientCIO

PORTFOLIO ATTRIBUTION REPORT

# Northstar Total Fund

Historical and governed ex ante attribution · prepared for the Investment Committee

HISTORICAL PERIOD
2025-07-31 to 2026-06-30

FORWARD VIEW AS OF
2026-10-08

BASE CURRENCY
USD

BENCHMARK ROLE
Policy

OVERALL VIEW
**Both lanes reconcile**

SCOPE REVIEWED
**5 of 5 · Complete**

# 01 Executive Attribution Summary

Realized and assumption-based effects

OVERALL VIEW

**Both lanes reconcile**

SCOPE REVIEWED · 5/5

ATTRIBUTION CONCLUSION

The portfolio's realized active return was -0.6pp, led by a -0.8pp public-equity selection effect S4. Governed ex ante attribution is +0.4pp on persisted one-year expected returns, with fixed-income selection the largest positive effect S5. Both lanes reconcile within their returned tolerance, but their realized linked and single-period expected bases are not interchangeable S4 S5.

| HISTORICAL ACTIVE RETURN | LARGEST HISTORICAL EFFECT | EX ANTE ACTIVE RETURN | LARGEST SUB-ALLOCATION EFFECT |
|---|---|---|---|
| -0.6pp | -0.8pp | +0.4pp | -0.4pp |
| 12 months to 2026-06-30 S4 | Public equity selection S4 | One-year expected basis S5 | US equity cumulative selection S4 |

AT A GLANCE

| AREA | EVIDENCE-BASED FINDING | STATUS |
|---|---|---|
| Historical result | Active return was -0.6pp; public-equity selection was the largest negative effect S4. | Available |
| Ex ante result | Expected active return is +0.4pp; fixed-income selection is the largest positive effect S5. | Available |
| Reconciliation | Both total effects reconcile to their returned active return within tolerance S4 S5. | Passed |
| Assumptions | Ex ante effects use persisted base-currency expected returns and current cohort weights S5. | Assumption |
| Coverage | All required sample sources are available S1 S2 S3 S4 S5. | Complete |

**Scope**

This is a monitoring analysis, not an investment recommendation or forecast. Ex ante expected returns and effects are assumptions, not guarantees.

**Key judgment — headline signal**

Observation: The sample report carries a reconciled-lanes signal versus the prior unresolved status S1. Why it matters: This signal focuses oversight on the report's most decision-relevant condition. Uncertainty: Evidence coverage may exclude newer or more detailed observations. What would change the view: A governed refresh showing a different signal or breached threshold.

**Key judgment — primary measure**

Observation: Historical active return is -0.6pp in the review evidence S2. Why it matters: This measure establishes the baseline for comparing later governed results. Uncertainty: The source basis may omit newer observations outside the reviewed evidence. What would change the view: A refreshed source report showing a change above 1%.

**Key judgment — monitoring priority**

Observation: US equity cumulative selection is the largest sub-allocation effect at -0.4pp S4. Why it matters: The depth-1 result identifies where the root detractor is concentrated. Uncertainty: The rerun covers only returned child allocations. What would change the view: A refreshed depth-1 result identifying another leading sub-allocation.

# 02 Benchmark and Data Basis

The two attribution lanes use different evidence

| | |
|---|---|
| Portfolio | **Northstar Total Fund** S2 |
| Period | **2025-07-31 to 2026-06-30** S4 |
| Parent cohort | **Root** S2 |
| Benchmark role | **Policy** S4 |
| Base currency | **USD** S2 |
| Historical basis | **Monthly total returns; beginning-of-period weights** S4 |
| Ex ante basis | **One-year expected returns; current normalized cohort weights** S5 |
| Data scope | **Illustrative sample data** S1 |

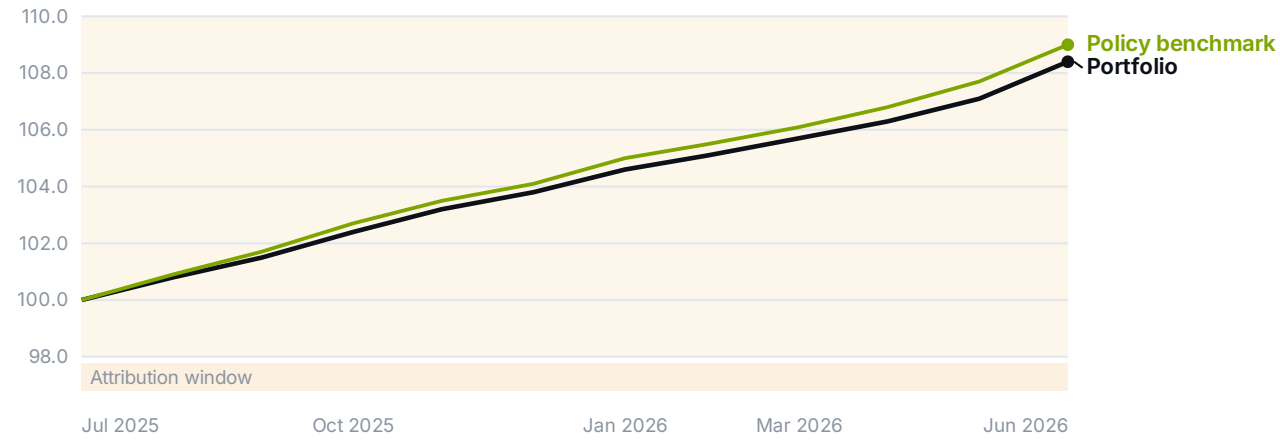
METHOD BOUNDARY

| LANE | WEIGHTS | RETURNS | HORIZON / LINKING | METHOD VERSION | SOURCE |
|---|---|---|---|---|---|
| Historical | Beginning of period | Realized total return | 12 monthly periods / symmetric Carino | realized_brinson_fachler | S4 |
| Ex ante | Current actual and persisted policy weights, normalized | Annualized expected return | One expected period / no linker | 1.0.0 | S5 |

## 03 Historical Returns Context

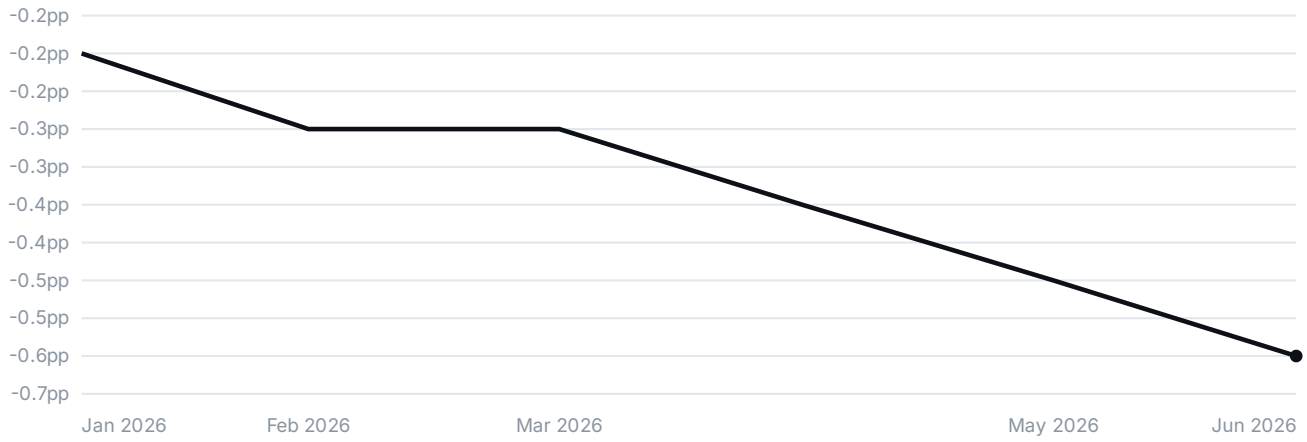
Observed performance provides context, not causal attribution

GROWTH OF 100




Full returned monthly series on the policy-benchmark entity basis S3.

ROLLING 12-MONTH EXCESS




Returned rolling 12-month arithmetic excess S3.

DRAWDOWN




Returned drawdown series S3.

STANDARD PERIODS TO 2026-06-30

| PERIOD | PORTFOLIO | BENCHMARK | EXCESS | COVERAGE |
|---|---|---|---|---|
| QTD | 2.8% | 3.2% | -0.4pp | Complete |
| YTD | 4.6% | 5.1% | -0.5pp | Complete |
| 1Y | 8.4% | 9.0% | -0.6pp | Complete |
| 3Y (ann.) | 7.1% | 7.5% | -0.4pp | Complete |

Returned labels, annualization and coverage are preserved S3.

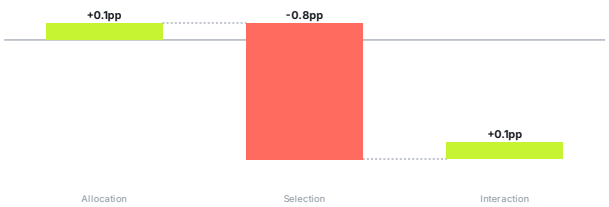
**Analysis — Relative returns trailed recently**

Observation: The portfolio trailed over all three periods: QTD, YTD and 1Y S3. Why it matters: The pattern establishes context for attribution results. Uncertainty: Aggregate return coverage does not identify causal effects. What would change the view: A later full-period report showing sustained positive excess return.

## 04 Historical Attribution

Realized Brinson-Fachler effects with symmetric-Carino linking

ALLOCATION → SELECTION → INTERACTION




HISTORICAL RECONCILIATION AND METHOD

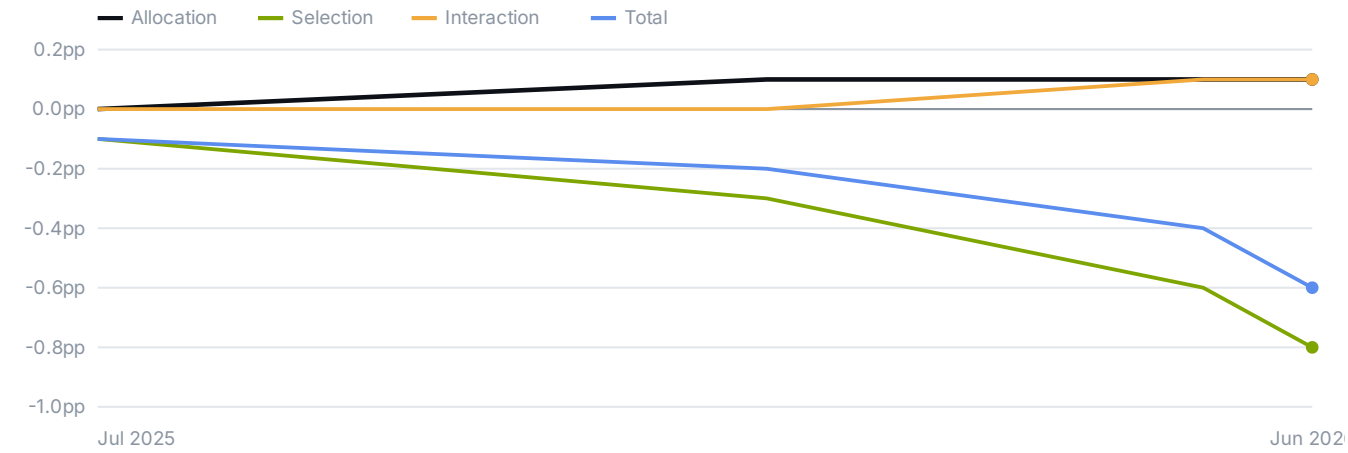
| | |
|---|---|
| Active return | **-0.6pp** S4 |
| Total effects | **-0.6pp** S4 |
| Method | **Realized Brinson-Fachler** S4 |
| Linking | **Symmetric Carino** S4 |
| Currency | **USD** S4 |

Returned linked effects S4.

HISTORICAL SEGMENT TOTAL EFFECTS



### CUMULATIVE LINKED EFFECTS



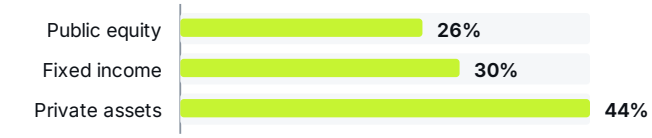

Returned cumulative symmetric-Carino linked effects S4.

### SEGMENT × MONTH LINKED TOTAL EFFECT

| | JUL 2025 | JAN 2026 | MAY 2026 | JUN 2026 |
|---|---|---|---|---|
| PUBLIC EQUITY | -0.1 | -0.2 | -0.2 | -0.1 |
| FIXED INCOME | 0.0 | 0.1 | 0.1 | 0.1 |
| PRIVATE ASSETS | 0.0 | -0.1 | -0.3 | 0.0 |

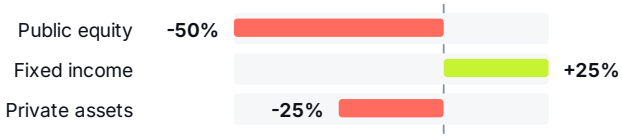
Returned monthly linked total effects by segment S4.

### CONTRIBUTION TO TRACKING ERROR



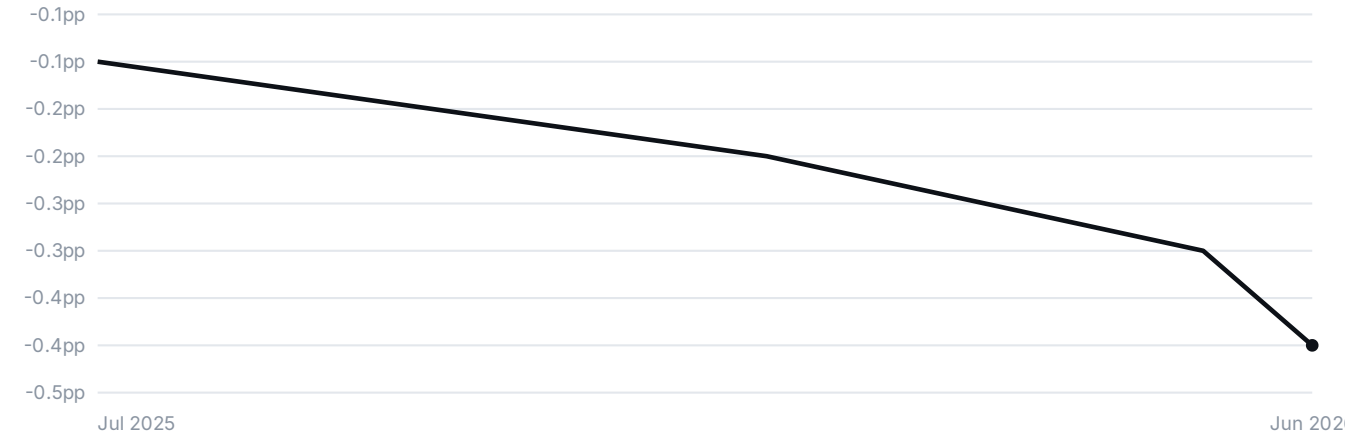

Returned active-risk contribution shares S4.

### RETURN EFFECT SHARE




Returned total effect divided by the sum of absolute returned segment effects S4.

### CUMULATIVE SELECTION INSIDE US EQUITY




Returned cumulative selection from the depth-1 attribution rerun S4.

### HISTORICAL TOTAL EFFECTS

| EFFECT | CONTRIBUTION | SOURCE |
|---|---|---|
| Allocation | +0.1pp | S4 |
| Selection | -0.8pp | S4 |
| Interaction | +0.1pp | S4 |
| Total attribution | -0.6pp | S4 |

HISTORICAL SEGMENT EFFECTS

| SEGMENT | ALLOCATION | SELECTION | INTERACTION | TOTAL |
|---|---|---|---|---|
| Public equity | +0.1pp | -0.8pp | +0.1pp | -0.6pp |
| Fixed income | +0.1pp | +0.2pp | 0.0pp | +0.3pp |
| Private assets | -0.1pp | -0.2pp | 0.0pp | -0.3pp |

Rows are returned effects and are not recalculated S4.

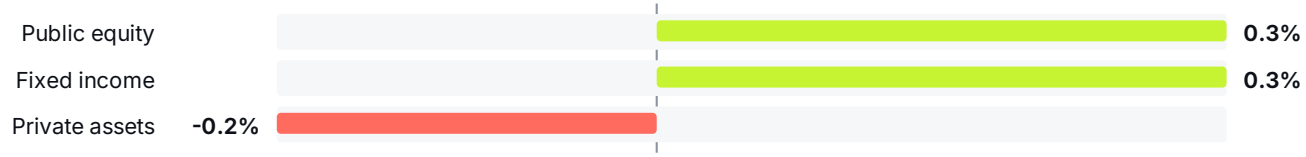
**Analysis — Equity selection drove detractor**

Observation: Public-equity selection is the largest detractor, and resolved managers have one open finding S4 S6. Why it matters: Effect and diligence evidence concentrate oversight there. Uncertainty: The balanced-regime model is contextual and not causal S7. What would change the view: Closed findings or refreshed results identifying another leading effect.

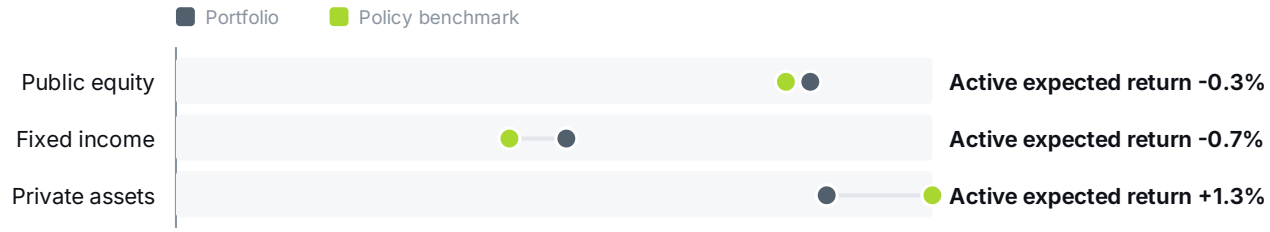
## 05 Governed Ex Ante Attribution

Single-period expected effects from persisted assumptions

GOVERNED EX ANTE SEGMENT TOTAL EFFECTS




EXPECTED RETURN BY SEGMENT




Returned portfolio and policy-benchmark expected returns S5.

**Assumptions, not forecasts**

The one-year expected returns and resulting effects are assumptions, not forecasts or guarantees S5.

EX ANTE TOTAL EFFECTS

| EFFECT | CONTRIBUTION | SOURCE |
|---|---|---|
| Allocation | +0.1pp | S5 |
| Selection | +0.2pp | S5 |
| Interaction | +0.1pp | S5 |
| Total attribution | +0.4pp | S5 |

EX ANTE RECONCILIATION AND METHOD

| | |
|---|---|
| Expected active return | **+0.4pp** S5 |
| Total effects | **+0.4pp** S5 |
| Method / version | **Expected Brinson-Fachler / 1.0.0** S5 |
| Horizon / linking | **One year / none** S5 |
| Return sources | **Allocation and benchmark expected return base** S5 |
| Assumption set / regime | **Not linked / Not linked** S5 |
| Currency | **USD** S5 |

EX ANTE SEGMENT EFFECTS

| SEGMENT | PORTFOLIO WT. | BENCHMARK WT. | PORTFOLIO EXP. RETURN | BENCHMARK EXP. RETURN | ALLOCATION | SELECTION | INTERACTION | TOTAL |
|---|---|---|---|---|---|---|---|---|
| Public equity | 55.0% | 50.0% | 7.8% | 7.5% | +0.1pp | +0.2pp | 0.0pp | +0.3pp |
| Fixed income | 30.0% | 35.0% | 4.8% | 4.1% | +0.1pp | +0.2pp | 0.0pp | +0.3pp |
| Private assets | 15.0% | 15.0% | 8.0% | 9.3% | 0.0pp | -0.2pp | 0.0pp | -0.2pp |

Inputs and effects are returned by the governed ex ante contract S5.

**Analysis — Assumption lineage remains incomplete**

Observation: Ex ante active return is +0.4pp, but no assumption set or regime is linked S5. Why it matters: Persisted inputs cannot be tied to a named release. Uncertainty: Input dates may differ by segment. What would change the view: A named release and regime record linked to every segment.

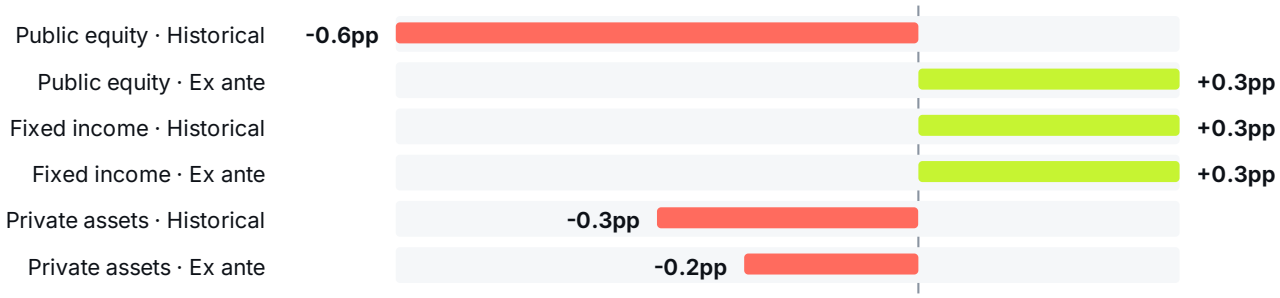
## 06 Historical-versus-Ex-Ante Scope and Comparison

Directional comparison with explicit basis separation

**Basis difference**

Historical effects use realized monthly observations and symmetric-Carino linking; ex ante effects use current normalized cohort weights and one expected period. The values are not directly interchangeable S4 S5.

HISTORICAL VS EX ANTE SEGMENT EFFECTS




Paired returned effects; lanes retain different bases S4 S5.

PUBLIC EQUITY
**Changed sign**
Historical – · ex ante + S4 S5

FIXED INCOME
**Same sign**
Historical + · ex ante + S4 S5

PRIVATE ASSETS
**Same sign**
Historical – · ex ante – S4 S5

DIRECTIONAL COMPARISON

| SEGMENT | HISTORICAL TOTAL EFFECT | EX ANTE TOTAL EFFECT | DIRECTION | BASIS NOTE |
|---|---|---|---|---|
| Public equity | -0.6pp | +0.3pp | Changed sign | Realized linked vs expected single period S4 S5 |
| Fixed income | +0.3pp | +0.3pp | Same sign | Realized linked vs expected single period S4 S5 |
| Private assets | -0.3pp | -0.2pp | Same sign | Realized linked vs expected single period S4 S5 |

**Analysis — Public equity changes sign**

Observation: Public equity is -0.6pp historically and +0.3pp ex ante S4 S5. Why it matters: The sign change separates experience from expectations. Uncertainty: The lanes use different periods and weight bases. What would change the view: Refreshed persisted results on the intended policy basis.

## 07 Diagnostics and Limitations

Reconciliation, coverage and model boundaries

HISTORICAL
**0.0 bp**
Tolerance 0.000001 bp · ✓ S4

EX ANTE
**0.0 bp**
Tolerance 0.000001 bp · ✓ S5

RECONCILIATION DIAGNOSTICS

| LANE | COVERAGE | RESIDUAL | TOLERANCE | WITHIN TOLERANCE | WARNINGS |
|---|---|---|---|---|---|
| Historical | Available | 0.0bp | 0.00000001pp | Yes | None S4 |
| Ex ante | Available | 0.0bp | 0.00000001pp | Yes | Assumption set and regime IDs are null S5 |

- Historical effects cover 12 complete monthly observations and use returned symmetric-Carino linking S4.
- Ex ante effects use current actual and persisted policy benchmark weights normalized within the root sibling cohort S5.
- Persisted expected returns are in USD; currency effects are not separately included S5.
- Assumption-set and regime identities are not attached to the persisted expectations S5.
- Missing evidence would remain unavailable rather than being treated as zero.

## 08 Coverage

Availability of every expected source

| | | |
|---|---|---|
| **Capabilities** | available | Portfolio Analytics live sample access S1 |
| **Portfolio and structure** | available | Root cohort; three segments S2 |
| **Historical returns context** | available | Complete through 2026-06-30 S3 |
| **Historical attribution** | available | 12 monthly periods; governed result available S4 |
| **Governed ex ante attribution** | available | One-year expected basis; governed result available S5 |
| **Largest-detractor manager findings** | available | One open finding for resolved public-equity managers S6 |
| **Macro regime** | available | Balanced regime context for the historical period S7 |

## A Appendix — Sources and methods

| TAG | EVIDENCE | BASIS | TOOL / VIEW | KEY PARAMETERS | AS OF | DATA SCOPE | VALIDATION | DIGEST |
|---|---|---|---|---|---|---|---|---|
| S1 | Capabilities | gradient_tool_metadata | get_gradient_capabilities | detail=full | 2026-10-08 | Illustrative sample data | passed | not_applicable_metadata_response |
| S2 | Portfolio and root cohort | gradient_tool | list_portfolios; get_portfolio_structure | view=allocation_tree | 2026-10-08 | Illustrative sample data | passed | sha256:222222222222 |
| S3 | Historical returns context | gradient_tool | get_portfolio_historical_returns | five sections | 2026-06-30 | Illustrative sample data | passed | sha256:333333333333 |
| S4 | Historical attribution | gradient_tool | get_portfolio_attribution | policy; root; all sections | 2026-06-30 | Illustrative sample data | passed | sha256:444444444444 |
| S5 | Governed ex ante attribution | gradient_tool | get_portfolio_ex_ante_attribution | policy; root; all sections | 2026-10-08 | Illustrative sample data | passed | sha256:555555555555 |
| S6 | Largest-detractor manager findings | gradient_tool | get_manager_diligence_findings | view=open; public-equity managers | 2026-06-30 | Illustrative sample data | passed | sha256:666666666666 |
| S7 | Macro regime | gradient_tool | get_macro_signals | view=gradient_signal; regime_state | 2026-06-30 | Illustrative sample data | passed | sha256:777777777777 |

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

Full digests — S2: sha256:2222222222222222222222222222222222222222222222222222222222222222; S3: sha256:3333333333333333333333333333333333333333333333333333333333333333; S4: sha256:4444444444444444444444444444444444444444444444444444444444444444; S5: sha256:5555555555555555555555555555555555555555555555555555555555555555; S6: sha256:6666666666666666666666666666666666666666666666666666666666666666; S7: sha256:7777777777777777777777777777777777777777777777777777777777777777

# B Appendix — Server Metric Methods and Disclosures

SERVER METRIC METHODS

| LANE | QUANTITY | METHOD / FORMULA | VERSION | BASIS / UNITS / TOLERANCE |
|---|---|---|---|---|
| Historical | Allocation, selection, interaction | Realized Brinson-Fachler with symmetric Carino | governed-portfolio-attribution-v1 | Monthly total return; decimal fraction; 1e-10 S4 |
| Ex ante | Allocation, selection, interaction | Single-period expected Brinson-Fachler | 1.0.0 | Annualized expected return; decimal fraction; 1e-10 S5 |
| Historical returns | Growth, rolling excess, drawdown | Server-returned indexed growth, rolling-window excess and peak-to-trough drawdown | governed-historical-returns-v1 | Monthly observations; index / percentage / percentage S3 |
| Historical active risk | Tracking-error contribution | Server-returned segment contribution to total active risk | governed-portfolio-attribution-v1 | Annualized tracking error; contribution share S4 |
| Historical periods | Cumulative and segment-month linked effects | Server-returned symmetric-Carino effects from periods and period segments | governed-portfolio-attribution-v1 | Monthly and cumulative percentage-point effects S4 |
| Historical drill-down | Largest sub-allocation cumulative selection | Server-returned depth-1 period selection effects without local reconstruction | governed-portfolio-attribution-v1 | Cumulative percentage-point selection effect S4 |
| Ex ante assumptions | Portfolio and benchmark expected return by segment | Server-returned expected returns applied to actual and policy segment weights | 1.0.0 | Annualized expected return; percentage S5 |
| Comparison | Segment effect pairing and sign agreement | Direct presentation of server-returned historical and ex ante segment effects | report-contract-v1 | Percentage-point effects; no cross-lane rebasing S4 S5 |

- Ex ante expected returns and effects are assumptions, not forecasts or guarantees.
- Historical attribution is realized and linked as returned.
- Historical and ex ante bases are not directly interchangeable.
- Missing evidence is not zero.
- AI-assisted monitoring analysis; not investment, legal or compliance advice.